

Lakshya Agarwal

☎ 408.410.3178 | ✉ lakshya6@illinois.edu | 📍 San Jose, California | 🔗 [LinkedIn](#) | 🐙 [GitHub](#)

EDUCATION

University of Illinois at Urbana-Champaign

Expected May 2028

Bachelor's Degree in Computer Engineering

GPA: 3.87/4

- **Coursework:** Data Structures & Algorithms, Applied Parallel Programming, Computer Systems & Programming

EXPERIENCE

Cisco

May 2026 – August 2026

Software Engineering Intern - Industrial IoT / AgenticOps

- Developed Cisco's **Agentic Runbook Creator (ARC)**, automating triage workflows for industrial network faults
- Built **SLM-driven** hypothesis generation & multi-stage runbook assembly from operator prompts & topology context
- Designed **privacy-preserving handoffs** obfuscating topology, names, IPs, & device IDs before cloud LLM analysis
- Created a **digital-twin evaluator** that injects faults into IOS-XE cloud simulations & validates runbook responses
- Orchestrated ARC's end-to-end workflow through **Flask**, allowing creation of expert-grade runbooks **on-premises**
- Cut network downtime from hours to **25-30 minutes** & scaled runbook coverage **beyond 13 predefined procedures**

Champaign Urbana Business and Engineering (CUBE) Consulting

Jan. 2026 – May 2026

Consultant for NuCurrent - Wireless Technology Startup

- Built a **React/FastAPI/SQL inventory platform** for NuCurrent, allowing for search, stock updates, & retrieval
- **Led frontend workflows** for fast search, item entry, check-in/check-out, barcode scanning, and label generation
- Designed a database schema & REST API for real-time inventory queries across **10 electronic component types**
- Implemented low-stock alert support & client handoff materials, enabling **maintainability & future expansion**

NeuroTechX @ UIUC

Aug. 2025 – Feb. 2026

Data Processing Engineer for Neurify

- Helped develop an **EEG-based emotion recognition headset** that classifies user emotions from brainwave data
- Built Python-based EEG processing workflows with Pandas and Matplotlib for real-time analysis & visualization
- Integrated the headset with Spotify Web API to **dynamically adjust Spotify music playback** based on emotion

PROJECTS

RepoFrame

June 2026 – August 2026

GitHub Repository Analyzer & Project Profile Generator

[Demo Website](#)

- Built a **Next.js/TypeScript & FastAPI/Python** tool turning GitHub repositories into evidence-backed project profiles
- Integrated **GitHub API** ingestion for repository metadata and recursive file trees with bounded source-file retrieval
- Implemented deterministic file ranking & multi-signal stack detection to construct inputs for **LLM-generated profiles**
- Built a bounded **Evidence Investigator** agent with allowlisted search and file-read tools to verify generated claims

QuackTA

April 2026

AI-Powered Interactive Study Companion - IEEE Bit-By-Bit Hackathon

- Developed a portable AI study companion utilizing **Raspberry Pi 5, Arduino Uno R4 Minima**, & an app-based UI
- Provided GPT-4o Mini with **UIUC ECE course and schedule context** for course-specific help & scheduling support
- Designed guided tutoring around **rubber-duck debugging** to help students reason through problems efficiently
- Implemented Arduino-controlled expressive bodily movement within a custom **3D-printed duck enclosure**

Lucky Arduino

Nov. 2023 – Aug. 2025

Electronics/Coding Content Creator

- Published **50+ Arduino/C++ tutorials** on [YouTube](#), generating **50K+ views & 200+ subscribers**
- Created a companion [personal website](#) with **500+ visits** to showcase projects & supplemental electronics content

SKILLS

Languages - Python (Pandas, Matplotlib, OpenCV, NumPy), C, C++, Java, JavaScript/TypeScript, HTML/CSS, SQL

Frameworks/Tools - FastAPI, Flask, React, Next.js, PostgreSQL, Supabase, Docker, Linux, Git, GitHub

AI/Agentic Systems - OpenAI API & Codex, Claude Code, Ollama, MCP, agentic workflows, structured outputs